OCTOBER 13, 2015

# MIDTERM I SOLUTION KEY

## FALL 2015

**1.** (10p) Let $f(x) = 1 + \cos x$ and $g(x) = x + \pi$, $0 \le x \le \pi$. Find a formula for the composite function $(f \circ g)(x)$, and determine its range. Sketch a graph of $(f \circ g)(x)$. Determine open interval(s) of $[0, \pi]$ where $(f \circ g)$ is increasing and where $(f \circ g)(x)$ is decreasing.

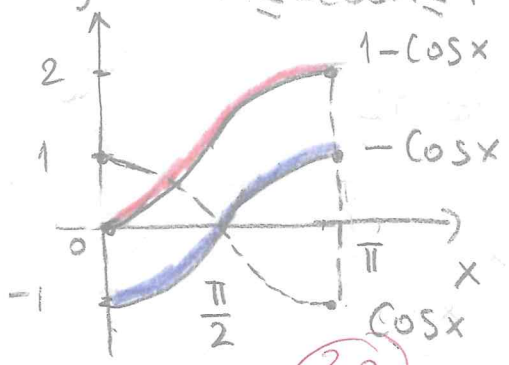
$$y = (f \circ g)(x) = f(g(x)) = f(x+\pi) = 1 + \cos(x+\pi) = 1 + \cos x \underbrace{\cos\pi}_{-1} - \sin x \underbrace{\sin\pi}_{0}$$

3p

$$(f \circ g)(x) = 1 - \cos x \; ; \quad -1 \le -\cos x \le 1 \rightarrow 0 \le \underbrace{1 - \cos x}_{y} \le 2$$

$R(f) = [0, 2]$

2p




3p

$f \circ g$ is increasing on $(0, \pi)$ 2p

**2.** (8p) Find each of the following limits (Limits should be calculated without the use of *L'Hôpital's Rule*!)

$$i)\ \lim_{h\to 0} \frac{\sqrt{2h+4} - 2}{h} = \lim_{h\to 0} \frac{\sqrt{2h+4} - 2}{h} \cdot \frac{\sqrt{2h+4} + 2}{\sqrt{2h+4} + 2}$$

4p

$$= \lim_{h\to 0} \frac{(2h + \cancel{4}) - \cancel{4}}{h(\sqrt{2h+4} + 2)} = \lim_{h\to 0} \frac{2\cancel{h}}{\cancel{h}(\sqrt{2h+4} + 2)} = \frac{2}{4} = \frac{1}{2}$$

$$ii)\ \lim_{x\to\infty} \left[\ln(1 + \frac{1}{x}) + e^{\frac{1}{x}}\right] = \lim_{x\to\infty} \ln(1 + \frac{1}{x}) + \lim_{x\to\infty} e^{\frac{1}{x}}$$

$$= \ln\left[\lim_{x\to\infty}(1 + \overset{0}{\cancel{\tfrac{1}{x}}})\right] + e^{\overset{0}{\lim_{x\to\infty} \frac{1}{x}}}$$

($\ln x$ and $e^x$ are continuous functions)

$$= \ln 1 + e^0 = 1$$

4p

**3.** (7p) Find a formula for the inverse of $f(x) = \ln(x-1)$. Determine the domain and the range of $f$ and $f^{-1}$.

$y = f(x) = \ln(x-1)$

2p $D(f)$: $x-1>0 \to (1,\infty)$

$R(f)$: $(-\infty,\infty)$

3p

S1. $y = \ln(x-1) \iff x-1 = e^y \to x = e^y+1$

S2. $y = f^{-1}(x) = 1+e^x$

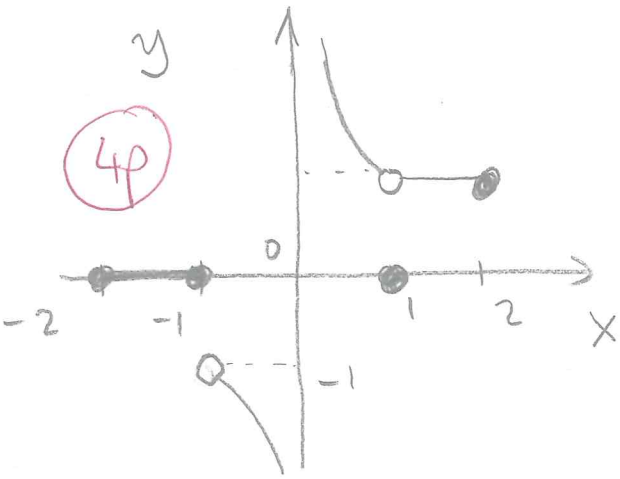
$D(f^{-1}) = (-\infty,\infty)$

$R(f) = (1,\infty)$ 2p

**4.** (16p) Graph the function

$$f(x) = \begin{cases} 0, & -2 \le x \le -1 \\ 1/x, & -1 < x < 1 \\ 0, & x = 1 \\ 1, & 1 < x \le 2 \end{cases}$$

Find right- and left- hand limits of $f$ at $x = -1, 0$, and 1. Determine if $f$ is *continuous* or *discontinuous* at those points, and identify the types of discontinuities (jump, removable or infinite discontinuities), if any.




4p

- $x=-1$: 4p

$\lim_{x\to -1^+} f(x) = -1$

$\lim_{x\to -1^-} f(x) = 0$

$\lim_{x\to -1} f(x)$ does not exist.

Conc: $f$ is not continuous at $x=-1$, and it is a jump discontinuity

- $x=0$: 4p

$f$ is not defined at $x=0$, and

$\lim_{x\to 0^+} f(x) = \infty$, $\lim_{x\to 0^-} f(x) = -\infty \to \lim_{x\to 0} f(x)$ does not exist

conc: $f$ is not continuous at $x=0$, and it is an infinite discontinuity.

- $x=1$: 4p

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x) = 1 \neq f(1) = 0$

conc: $f$ is discontinuous at $x=1$, and it is a removeble discontinuity

**5.** (8p) Show that the equation $x + 2\cos x = 0$ has at least one solution. 2p

The function $f(x) = x + 2\cos x$ is continuous on any $[a,b]$.

$f(-\frac{\pi}{2}) = -\frac{\pi}{2} + 0 = -\frac{\pi}{2} < 0$ ; $f(0) = 0 + 2 > 0$ 4p

By the Intermediate Value Theorem, there is a number $c$ in $[-\frac{\pi}{2}, 0]$ such that $f(c) = 0 \rightarrow c + 2\cos c = 0$ 2p

**6.** (12p) Find the *asymptotes* of the graph of $f(x) = \dfrac{1-2x}{2x-4}$. Determine the behavior of the function near the vertical asymptote by evaluating the right-hand and left-hand limits of $f(x)$. Then sketch a rough graph of $f$.

HA: $\lim\limits_{x \to \mp\infty} \dfrac{1-2x}{2x-4} = \lim\limits_{x \to \mp\infty} \dfrac{\cancel{x}(\overset{0}{\frac{1}{x}} - 2)}{\cancel{x}(2 - \frac{4}{\cancel{x}})} = -\frac{2}{2} = -1 \rightarrow$ 4p

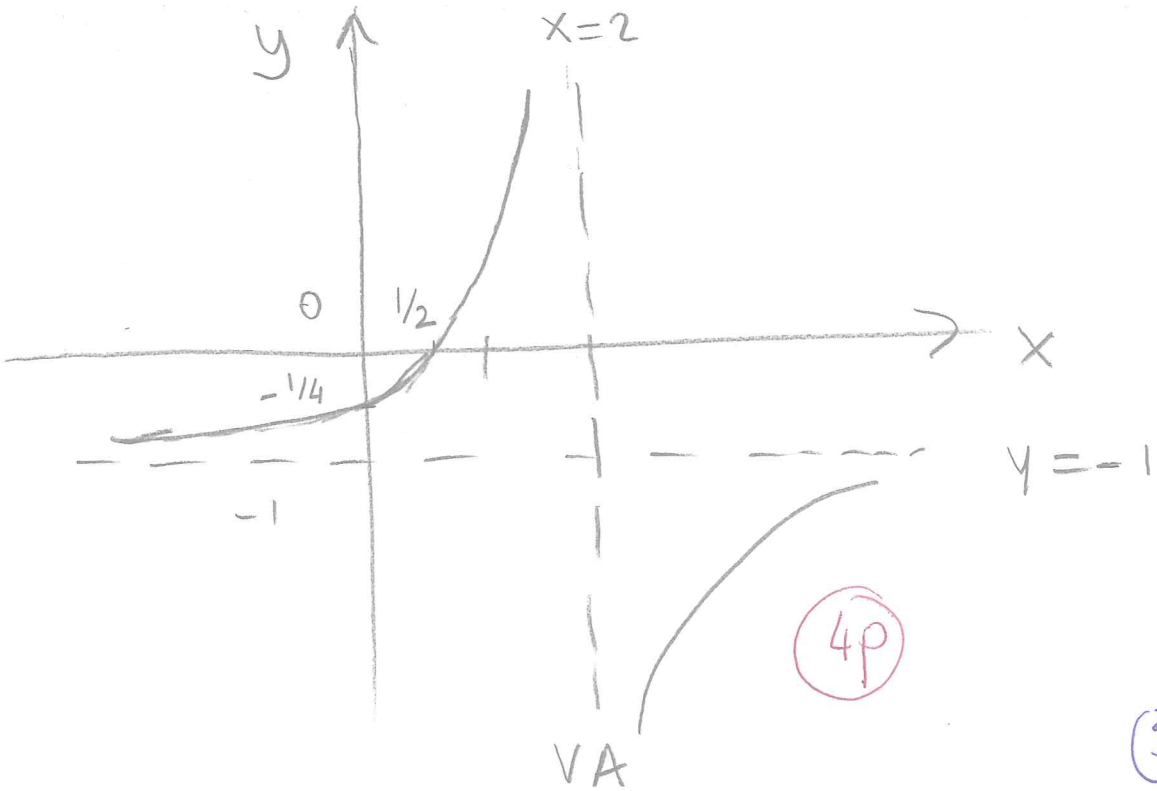
$y = -1$ is the HA

VA at $x = 2$:

$\lim\limits_{x \to 2^+} \dfrac{\overset{<0}{1-2x}}{\underset{>0}{2x-4}} = -\infty$ $\qquad$ $\lim\limits_{x \to 2^-} \dfrac{\overset{<0}{1-2x}}{\underset{<0}{2x-4}} = +\infty$ 4p

$f(0) = -\frac{1}{4}$

$f(\frac{1}{2}) = 0$




4p

3

7. (7p) Using the limit definition of the derivatives find $f'(x)$ if $f(x) = \dfrac{1}{2x+1}$ (no partial credits for direct calculation!)

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\frac{1}{2(x+h)+1} - \frac{1}{2x+1}}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{(2x+1)} - [2(\cancel{x}+h)+\cancel{1}]}{h(2x+1)[2(x+h)+1]} = \lim_{h\to 0} \frac{-2\cancel{h}}{\cancel{h}(2x+1)[2(x+h)+1]}$$

$\downarrow$ cont. f.

$$f'(x) = \frac{-2}{(2x+1)^2}$$ 7p

8. (10p) Suppose that the function $f(x)$ and its derivative $f'(x)$ have the following values at $x = 0$ and $x = 1$.

| $x$ | $f(x)$ | $f'(x)$ |
|---|---|---|
| 0 | 9 | -2 |
| 1 | -3 | 1 |

Find $F'(1)$ and $G'(0)$, where $F(x) = \sqrt{x}f(x)$ and $G(x) = \dfrac{f(x)}{2+\cos x}$.

$$F'(x) = \frac{1}{2\sqrt{x}} f(x) + \sqrt{x}\, f'(x)$$ (Product Rule) 3p

$$F'(1) = \frac{1}{2} f(1) + f'(1) = \frac{-3}{2} + 1 = -\frac{1}{2}$$ 2p

$$G'(x) = \frac{f'(x)(2+\cos x) + f(x)\sin x}{(2+\cos x)^2}$$ (Quotient Rule) 3p

$$G'(0) = \frac{f'(0)(2+1) + f(0).0}{9} = \frac{1}{3}(-2) = \frac{-2}{3}$$ 2p

4

**9.** (8p) Does the curve $f(x) = \dfrac{8}{x^2+4}$ have any horizontal tangent? If so, write an equation for the horizontal tangent line.

4p

$$f'(x) = \frac{0(x^2+4) - 8 \cdot 2x}{(x^2+4)^2} = \frac{-16x}{(x^2+4)^2} = 0 \to f \text{ has a HT at } x=0:$$

$m = f'(0) = 0$

Eq. for the TL through $(0,2)$ with slope $m=0$ is

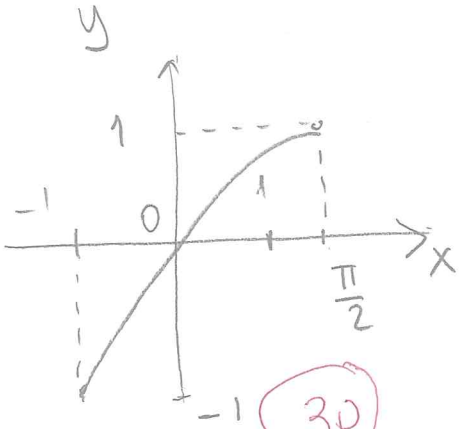
$y - 2 = 0(x-0)$

$y = 2$ 4p

**10.** (14p) Graph the function

$$f(x) = \begin{cases} x, & -1 \le x < 0 \\ \sin x, & 0 \le x \le \frac{\pi}{2} \end{cases}$$

*i)* Use continuity test to determine whether $f$ is continuous or discontinuous at $x = 0$.
*ii)* Compute the right-hand and left-hand derivatives as limits to show whether $f$ is differentiable or not differentiable at $x = 0$.




3p

i) Continuity test:

$f(0) = \sin 0 = 0$ exists

$\lim_{x \to 0} f(x) = 0$ exists

$\lim_{x \to 0} f(x) = 0 = f(0)$

f is cont. at $x=0$

4p

ii) RHD: $\lim_{h \to 0^+} \dfrac{f(0+h) - f(0)}{h} = \lim_{h \to 0^+} \dfrac{\sin h - 0}{h} = 1$ 3p

LHD: $\lim_{h \to 0^-} \dfrac{f(0+h) - f(0)}{h} = \lim_{h \to 0^-} \dfrac{h - 0}{h} = 1$ 3p

$f'(0) = 1$ exists since RHD = LHD. Thus

f is differentiable at $x = 1$ 1p